Two identities for cotangent line classes

Proofs of Zvonkine's Propositions 2.17 and 2.18

Lecture notes (about 35–45 minutes)

The two ideas. The normal bundle to a marked section is the *tangent* line at that mark, so its first Chern class is $-\psi_i$. When a new marked point collides with the ith mark, stable reduction inserts a three-pointed rational bubble; this changes the cotangent line by one copy of the bubbling divisor. The local equation behind the latter statement is simply $z = tu$.

1 Notation and statements

Fix g, n with $2g - 2 + n > 0$ and work in rational cohomology or the rational Chow ring of the smooth Deligne–Mumford stacks. Let

$$B = \overline{\mathcal{M}}_{g,n}, \qquad X = \overline{\mathcal{M}}_{g,n+1}, \qquad p : X \longrightarrow B$$

be the morphism forgetting the last mark and stabilizing. Recall the natural identification $X \simeq \overline{\mathcal{C}}_{g,n}$: the forgetful morphism *is* the universal curve.

The cotangent line at the ith mark is a line bundle on each moduli space. We distinguish them by superscripts:

$$L_i^{(n)} \longrightarrow B, \qquad L_i^{(n+1)} \longrightarrow X, \qquad \psi_i^{(n)} = c_1(L_i^{(n)}), \quad \psi_i^{(n+1)} = c_1(L_i^{(n+1)}).$$

Write $\sigma_i : B \to X \simeq \overline{\mathcal{C}}_{g,n}$ for the ith section and $D_i = \sigma_i(B)$ for its divisor. Viewed instead as a divisor of $\overline{\mathcal{M}}_{g,n+1}$, it is

$$\Delta_i = \delta_{0,\{i,n+1\}} :$$

the locus of stable curves with a rational tail carrying precisely marks $i, n+1$ and one attaching node. Thus $[D_i] = [\Delta_i]$ in $H^2(X;\mathbb{Q})$, and $p|_{D_i}$ is an isomorphism onto B.

Proposition 1 (Zvonkine 2.17: forgetting a marked point)**.** *For $1 \le i \le n$,*

$$\boxed{\psi_i^{(n+1)} = p^*\psi_i^{(n)} + [\Delta_i].} \tag{1}$$

In fact there is a natural line-bundle isomorphism

$$L_i^{(n+1)} \simeq p^*L_i^{(n)} \otimes \mathcal{O}_X(\Delta_i). \tag{2}$$

Proposition 2 (Zvonkine 2.18: self-intersection of a section)**.** *For every integer $k \ge 0$,*

$$\boxed{p_*\big([D_i]^{k+1}\big) = (-\psi_i^{(n)})^k.} \tag{3}$$

2 Proposition 2.18: the section and its normal bundle

It is best to prove Proposition 2 first. Near the image D_i of the section σ_i, the map p is smooth: by definition, a marked point cannot be a node. Its relative cotangent bundle is therefore locally a line bundle, and

$$L_i^{(n)} = \sigma_i^*\omega_p,$$

where ω_p denotes the relative dualizing sheaf (equal to the usual relative cotangent line in the smooth locus).

Lemma 1. *The normal bundle of D_i in X is*

$$N_{D_i/X} \simeq (L_i^{(n)})^\vee.$$

Consequently, $\sigma_i^*[D_i] = -\psi_i^{(n)}$.

Proof. Since $p \circ \sigma_i = \mathrm{id}_B$, the tangent map of p splits along the section. The normal direction to D_i is exactly the vertical tangent direction:

$$N_{D_i/X} \simeq \sigma_i^* T_{X/B} \simeq (\sigma_i^* \omega_p)^\vee = (L_i^{(n)})^\vee.$$

Equivalently, at a pointed curve $(C; x_1, \dots, x_n)$ the normal direction is $T_{x_i}C$, whereas the fiber of $L_i^{(n)}$ is $T_{x_i}^*C$. Finally, for any smooth divisor $D \subset X$,

$$\mathcal{O}_X(D)|_D \simeq N_{D/X},$$

so $\sigma_i^*[D_i] = c_1(N_{D_i/X}) = -\psi_i^{(n)}$. □

Proof of Proposition 2. The self-intersection (Gysin) formula for the divisor inclusion σ_i states that for any $\alpha \in H^*(X)$,

$$[D_i]\,\alpha = \sigma_{i*}\sigma_i^*\alpha.$$

Taking $\alpha = [D_i]^k$ and using Lemma 1,

$$[D_i]^{k+1} = \sigma_{i*}\big((\sigma_i^*[D_i])^k\big) = \sigma_{i*}\big((-\psi_i^{(n)})^k\big).$$

Apply p_* and use $p \circ \sigma_i = \mathrm{id}_B$:

$$p_*([D_i]^{k+1}) = (p \circ \sigma_i)_*((-\psi_i^{(n)})^k) = (-\psi_i^{(n)})^k.$$

This includes $k = 0$, for which $p_*[D_i] = 1$. □

Classroom check (genus zero). For $g = 0, n = 4$, the surface $\overline{\mathcal{M}}_{0,5}$ is the universal curve over $\overline{\mathcal{M}}_{0,4} \simeq \mathbb{P}^1$. Since $\int_{\overline{\mathcal{M}}_{0,4}} \psi_i = 1$,

$$\int_{\overline{\mathcal{M}}_{0,5}} D_i^2 = -1.$$

Thus the marked section is a (-1)-curve in the smooth surface $\overline{\mathcal{M}}_{0,5}$. In this example one can actually *see* the normal-bundle sign.

3 Proposition 2.17: collision of markings and the bubbling divisor

We now prove Proposition 1. The issue is that there are *two different* cotangent lines: one on the original $(n+1)$-pointed curve and one after forgetting the last point.

Step 1: A natural comparison map

Given $(C; x_1, \dots, x_{n+1}) \in X$, let

$$\rho : C \longrightarrow C'$$

be the stabilization map after forgetting x_{n+1}. More precisely, on X there are two families: the universal $(n+1)$-pointed family and the pullback of the universal n-pointed family along p. The stabilization is a morphism between these families. Its pullback map on relative dualizing differentials, restricted to the ith sections, gives a natural homomorphism

$$\Phi : p^* L_i^{(n)} \longrightarrow L_i^{(n+1)}. \tag{4}$$

Away from Δ_i, the contraction ρ is an isomorphism near the ith mark, hence Φ is an isomorphism there.

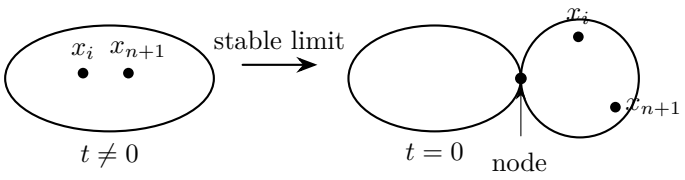
Why is Δ_i the only exceptional divisor? Forgetting a single marking can contract a rational component with only two special points remaining. The contraction changes the neighborhood of x_i exactly when that component contains both x_i and x_{n+1} and meets the rest of the curve in one node. This is the defining situation for Δ_i. A rational bridge with x_{n+1} and two nodes may also be contracted, but this does not change the neighborhood of any surviving marked point.

Step 2: A transverse one-parameter collision

Work near a general point of Δ_i. Keep the underlying smooth n-pointed curve fixed and choose a local coordinate z centered at x_i. Let the new marking move by

$$x_i : z = 0, \qquad x_{n+1} : z = t.$$

For $t \neq 0$ these are two different marks. At $t = 0$ they collide, so the stable limit requires a rational bubble.



Blow up the point $(z, t) = (0, 0)$ in the total space of the family. In the chart containing both marked points, the blowup is described by

$$\boxed{z = tu}, \tag{5}$$

where (u, t) are the new coordinates. The proper transforms of the two sections become

$$x_i : u = 0, \qquad x_{n+1} : u = 1.$$

At $t = 0$, the exceptional divisor is the rational tail with coordinate u. The attaching node is at $u = \infty$. (In the other blowup chart, with $t = zv$, the central fiber is the nodal curve $zv = 0$.)

Step 3: Compare cotangent vectors

On the original curve at x_i, a local generator of $L_i^{(n)}$ is dz. On the bubbled curve at the same marked point, a local generator of $L_i^{(n+1)}$ is du.

Differentiate $z = tu$:

$$dz = t\,du + u\,dt.$$

For *relative* differentials along a fiber of the family, $dt = 0$. Hence the comparison map (4) is represented locally by

$$\boxed{dz \longmapsto t\,du.} \tag{6}$$

It is invertible for $t \neq 0$; at $t = 0$ it has a *simple zero*. The transverse parameter t is a local equation of Δ_i, so

$$\operatorname{div}(\Phi) = \Delta_i$$

with multiplicity exactly one.

Step 4: The divisor of the comparison map

A nonzero homomorphism $E \to F$ of line bundles is a section of $F \otimes E^{-1}$. If its zero divisor is D, then $F \otimes E^{-1} \simeq \mathcal{O}(D)$.

Apply this to (4) and (6). We obtain

$$L_i^{(n+1)} \otimes (p^* L_i^{(n)})^{-1} \simeq \mathcal{O}_X(\Delta_i),$$

which is (2). Taking first Chern classes proves (1).

Remark (Why the local calculation proves the global statement). The comparison map is an isomorphism outside Δ_i. At a generic point of this irreducible divisor, the blowup chart shows vanishing to order one. On the smooth moduli stack, a nonzero map between line bundles defines an effective Cartier divisor, determined by its codimension-one multiplicities. More complicated intersections of boundary strata, which have codimension at least two here, give no additional divisor terms. The argument also works on local orbifold charts.

4 Compatibility: the two propositions describe the same geometry

The divisor Δ_i is isomorphic to B: we attach the unique three-pointed rational curve to the ith marking. Thus $L_i^{(n+1)}|_{\Delta_i}$ is trivial, because the marked point i lies on the rigid three-pointed rational component:

$$L_i^{(n+1)}|_{\Delta_i} \simeq \mathcal{O}_{\Delta_i}, \qquad \psi_i^{(n+1)} \cdot [\Delta_i] = 0. \tag{7}$$

Restrict (2) to Δ_i. Using $\mathcal{O}_X(\Delta_i)|_{\Delta_i} \simeq N_{\Delta_i/X}$ yields

$$\mathcal{O}_{\Delta_i} \simeq L_i^{(n)} \otimes N_{\Delta_i/X}, \qquad N_{\Delta_i/X} \simeq (L_i^{(n)})^\vee,$$

just as in Lemma 1. In cohomology,

$$[\Delta_i]^2 = -p^* \psi_i^{(n)} \cdot [\Delta_i]. \tag{8}$$

Pushing (8) forward gives $p_*[\Delta_i]^2 = -\psi_i^{(n)}$.

Useful identity. For every $d \geq 1$,

$$(\psi_i^{(n+1)})^d = p^*((\psi_i^{(n)})^d) + p^*((\psi_i^{(n)})^{d-1}) \cdot [\Delta_i]. \tag{9}$$

To verify this, expand $(p^*\psi_i^{(n)} + [\Delta_i])^d$ and repeatedly use (8) (or use $\psi_i^{(n+1)}[\Delta_i] = 0$). This formula is the workhorse in proofs of the string equation.

5 Optional application: the string equation

For different i, j, the divisors Δ_i and Δ_j are disjoint: the single new mark $n+1$ cannot belong to two distinct three-pointed rational tails. Hence

$$[\Delta_i][\Delta_j] = 0 \quad (i \neq j), \qquad p_*[\Delta_i] = 1, \qquad p_*1 = 0.$$

For nonnegative integers $d_1, \ldots, d_n$, expand the product using (9) (for $d_i = 0$, simply use $(\psi_i^{(n+1)})^0 = 1$). Mixed terms involving two different Δ_i vanish, and p_* annihilates the term pulled back entirely from B. The result is

$$p_*\left(\prod_{i=1}^{n}(\psi_i^{(n+1)})^{d_i}\right) = \sum_{\substack{1\leq j\leq n\\ d_j>0}} (\psi_j^{(n)})^{d_j-1} \prod_{i\neq j}(\psi_i^{(n)})^{d_i}. \tag{10}$$

After integration over $\overline{\mathcal{M}}_{g,n}$, this becomes

$$\left\langle \tau_0 \prod_{i=1}^{n} \tau_{d_i} \right\rangle_g = \sum_{\substack{1\leq j\leq n\\ d_j>0}} \left\langle \tau_{d_j-1} \prod_{i\neq j} \tau_{d_i} \right\rangle_g .$$

This is the string equation.

Board summary

Normal bundle:	$N_{D_i/X} \simeq (L_i^{(n)})^\vee$,
Self-intersection:	$p_*([D_i]^{k+1}) = (-\psi_i^{(n)})^k$,
Collision chart:	$z = tu, \qquad dz \equiv t\,du \pmod{dt}$,
Forgetting formula:	$\psi_i^{(n+1)} = p^*\psi_i^{(n)} + [\Delta_i]$.

Reference. D. Zvonkine, "An introduction to moduli spaces of curves and their intersection theory," §2.2.1, Propositions 2.17–2.18 (p. 688). See also §3.3.1 for the identification of the boundary divisor with the marked section and §4.1 for the string equation.